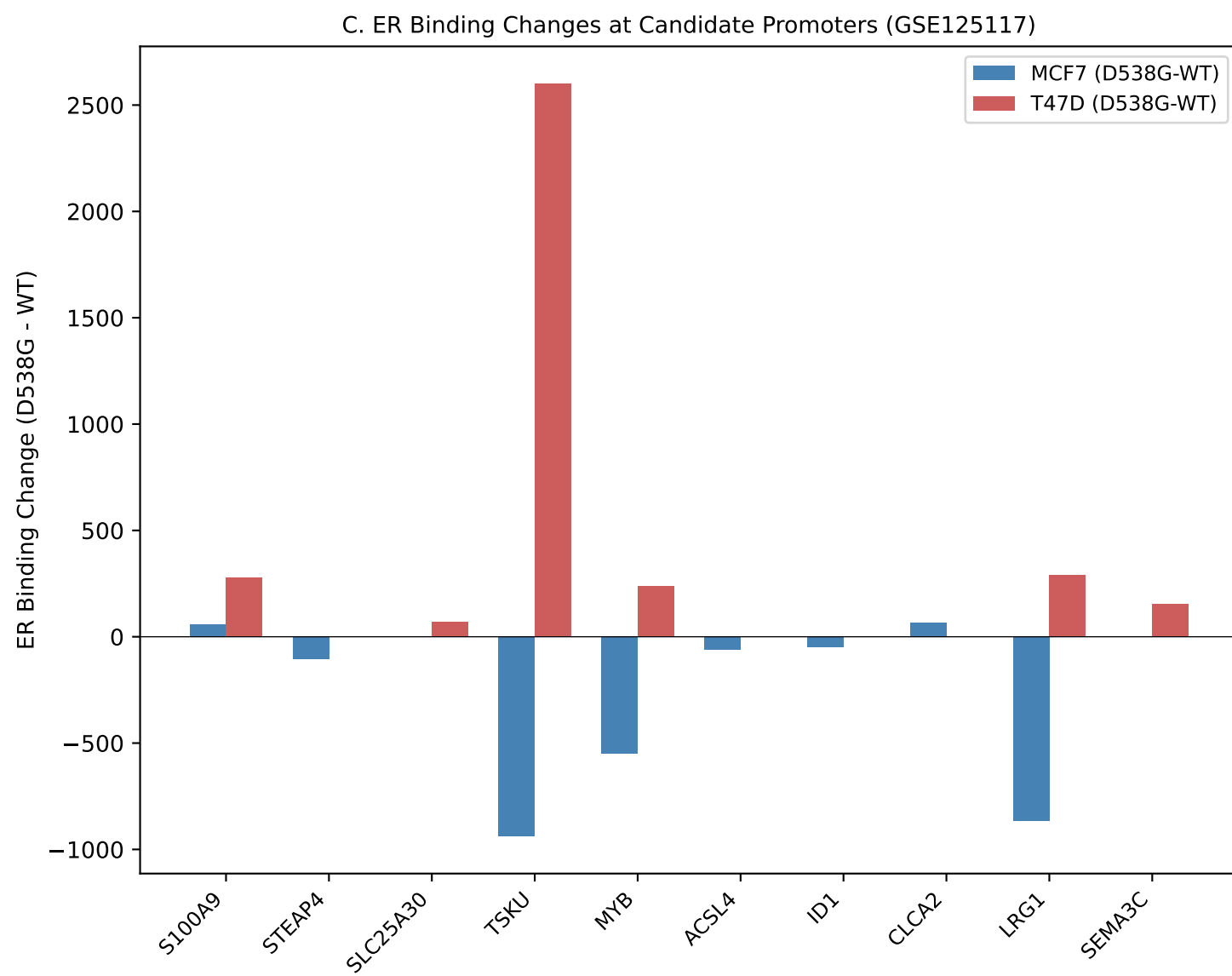
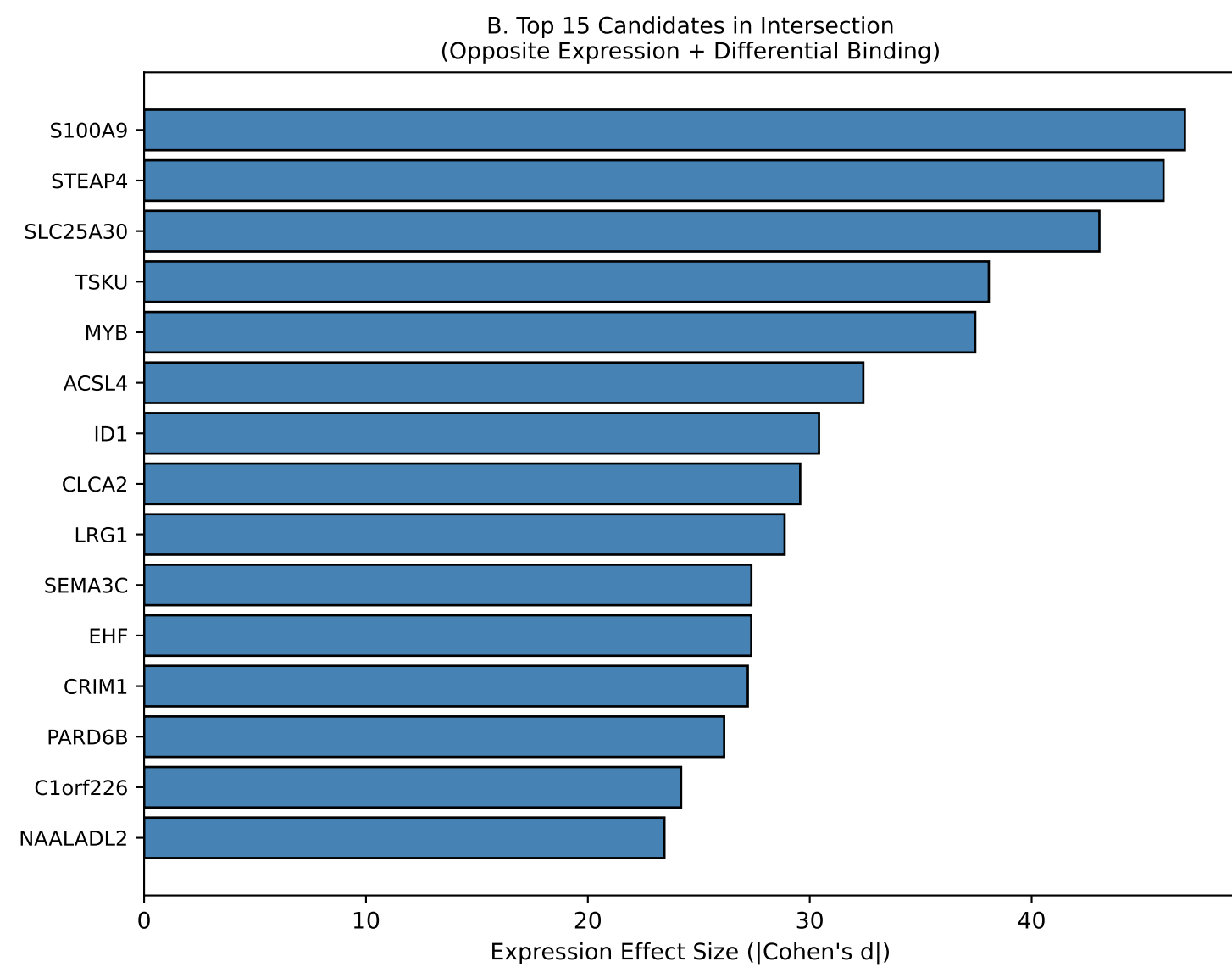
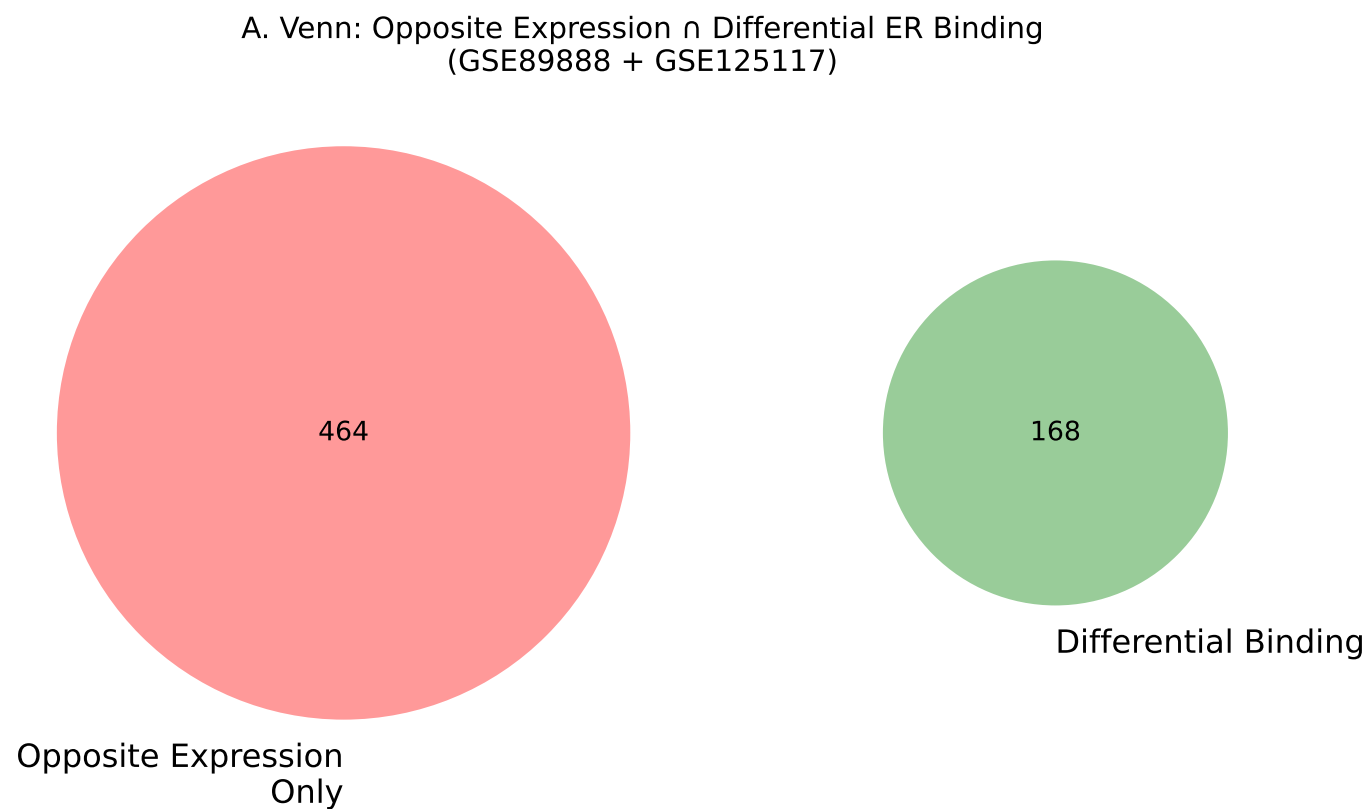




VENN DIAGRAM APPROACH: WINNOWING CANDIDATES

STARTING POINT:

- 632 genes with opposite expression (MCF7 vs T47D response to D538G)

FILTER BY ER BINDING:

- 171 have any ER binding at promoter
- 168 have DIFFERENTIAL ER binding (D538G changes binding vs WT)

INTERSECTION:

- 168 genes have BOTH:
 - Opposite expression regulation
 - Differential ER binding at promoter

HSP90B1 STATUS:

- In intersection: YES
- Rank in intersection: #36/168

INTERPRETATION:

The intersection contains genes where:

- Expression is opposite in MCF7 vs T47D
- ER binding changes upon D538G mutation

These are the strongest candidates for ER-mediated regulation of the phenotype.